

BFF5915 - Options, futures and risk management

Lecture 1

- Introduction to derivatives:
- Forward and futures contracts

* Throughout the semester, all predictions are hypothetical examples although they might be set up based on real –world data. In general, teaching materials are not investment advice.

Why do derivatives exist?

A world full of risk

- Fluctuating **gold prices** create uncertainty for mining companies about sale prices.
- Changing **exchange rates** worry Australian importers billed in USD.
- **Stock market volatility** raises concerns for fund managers about potential crashes.
- The risk of rising **interest rates** alarm companies planning to borrow soon.
- Unusual **weather** threatens an orange farmer's crop yield.
- **Derivative securities help manage these risks.**

What is a derivative?

- **A derivative security is a financial instrument whose value depends on (“derives from”) another variable.**
 - This variable could be the **price of assets** like stocks, interest rates, or exchange rates.
 - It might be the **price of commodities** such as gold, oil, coca or orange juice.
 - Or it could be a **non-price variable**, like temperature, rainfall, or carbon emissions.
- **Common derivatives include forwards, futures, options, and swaps.**

Derivatives Are Enormous in Size

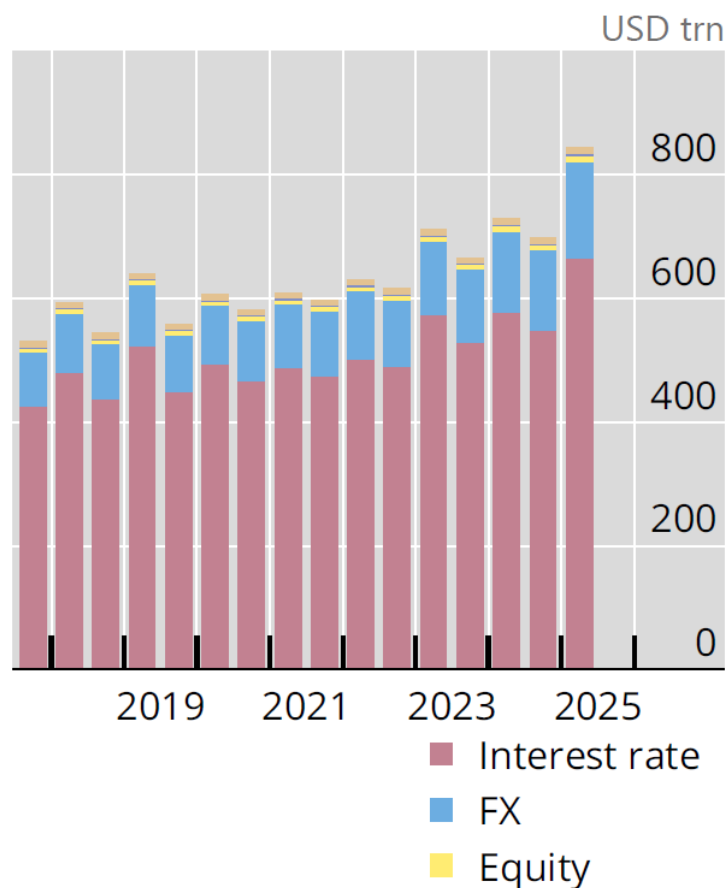
- <https://www.visualcapitalist.com/all-of-the-worlds-money-and-markets-in-one-visualization-2022/>
- Since options were first traded in Chicago in 1973, the use and trading of derivatives has exploded.
- The value of derivative trading dwarfs the value of trading in the underlying securities.

Derivatives Are Enormous in Size

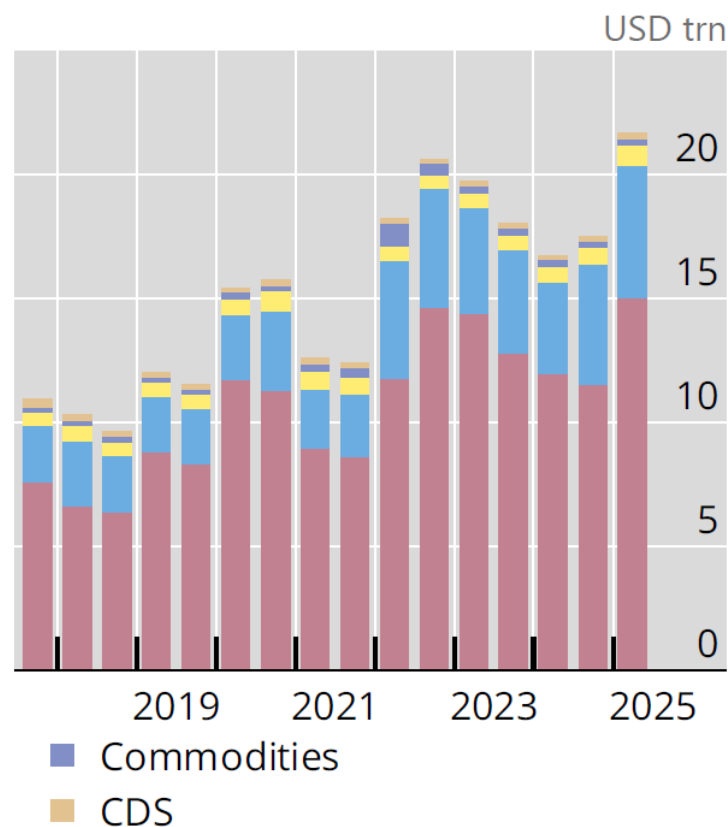
- Modern risk management depends heavily on derivatives
 - so does structuring virtually all major financial deals
 - so do many other areas in business, such as executive compensation.
- Basic familiarity with derivatives is essential for understanding how finance is being done today in real world firms and markets.

Derivatives Are Enormous in Size

Notional principal



Gross market value



Source: BIS OTC derivatives statistics.

In addition to OTC markets, **exchange-traded derivatives** — like futures and options on regulated exchanges — also represent **\$50-60 trillion in notional exposure**.

Risk management using derivatives

- Firms are exposed to risks **as part of their activity**:
 - Airlines exposed to fuel prices, multinationals exposed to FX fluctuations, banks are exposed to interest rate risk, etc.
- **Risk management** is about **identifying exposures** and **controlling their impact** on firm value.
- Derivatives allow firms to **adjust risk without altering core operations**.
 - Hedge unwanted exposures and stabilize cash flows $\Rightarrow$ reducing chances of financial distress.
 - Reduce earnings volatility $\Rightarrow$ smooth taxable income, stable dividend policy.
 - Improve financial planning $\Rightarrow$ better investment options.
 - Protect balance sheets $\Rightarrow$ prevent costly recapitalization.

Example of risk management by Qantas

20 FINANCIAL RISK MANAGEMENT

(A) RISKS

The Qantas Group is subject to financial risks which are an inherent part of the operations of an airline. The Qantas Group manages these risk exposures using various financial instruments, governed by a set of policies approved by the Board. The Qantas Group's policy is not to enter into, issue or hold derivative financial instruments for speculative trading purposes.

The Qantas Group uses different methods to assess and manage different types of financial risk to which it is exposed. These methods include correlations between risk types, sensitivity analysis in the case of interest rate, foreign exchange and other price risks, and ageing analysis and sensitivity analysis for liquidity and credit risk. A summary of these risks has been presented below:

Risk	Nature of Risk	Management of Risk
Liquidity risk	Difficulty in meeting financial liability obligations.	Remaining within optimal capital structure, targeting a minimum liquidity level, ensuring long-term commitments are managed, maintaining access to a variety of additional funding sources and managing maturity profiles.
Interest rate risk	Fluctuations in the fair value or future cash flows of a financial instrument because of changes in market interest rates.	Floating versus fixed rate debt framework, interest rate swaps, forward rate agreements and options.
Foreign exchange risk	Fluctuations in the fair value of future cash flows or assets/liabilities denominated in a currency other than AUD because of changes in foreign exchange rates.	Forward foreign exchange contracts, currency options, cross-currency swaps and fair value designation of non-derivative foreign currency assets and liabilities in a hedge relationship.
Fuel price risk	Exposure of future AUD fuel to unfavourable USD-denominated price movements and foreign exchange movements.	USD price – options and swaps on jet kerosene, gasoil and crude oil. Foreign exchange risk – foreign exchange contracts and currency options.
Credit risk	Potential loss from a transaction in the event of a default by a counterparty during the term or on settlement of a transaction.	Trade Debtor counterparties – stringent credit policies and accreditation of travel agents through industry programs. Other financial asset counterparties – transact only with counterparties that have acceptable credit ratings and counterparty limits.

Economic Function of Derivatives

1. Risk Transfer

- They shift risk from agents who do not want it to agents who are willing to bear it.

2. Risk Allocation

- They allow risks to be distributed to those with:
- Better information
- Greater risk tolerance
- Better diversification capacity

3. Market Efficiency

- Derivative prices incorporate expectations about market volatility, interest rates, default risk, and future prices.
- They often lead the underlying market in information processing.

Do derivatives have a dark side?

- Derivative securities are often complicated, and many people do not properly understand them.
- The use of derivatives can increase or decrease the risk of a certain portfolio.
- It can be used for taking **a lot of** risk.
- Derivatives markets have not always been tightly regulated.
- As a result, Derivatives have occasionally been at the center of **high-profile scandals** resulting in huge losses.

Long-Term Capital Management (LTCM)

- Founded in 1994 . Highly leveraged hedge fund run by elite financial experts, including Myron Scholes and Robert C. Merton (which we will meet later!)
- Engaged in complex arbitrage strategies across global markets, using instruments such as options and swaps.
- LTCM used massive leverage (30:1 or more) to profit from small market inefficiencies.
- 1998 Russian default triggered a global flight to quality. Spreads widened.
- Losses exceeded **\$4.6 billion** in a matter of weeks, threatening to default on **over \$1 trillion in notional derivative exposure**.
- Exposures were large enough to risk collapse of major counterparties (e.g., Merrill, Goldman).
- **Federal Reserve organized a \$3.6B private bailout** by major banks in September 1998. It was dissolved in 2000.
- Recommended reading: “When Genius Failed” by Roger Lowenstein

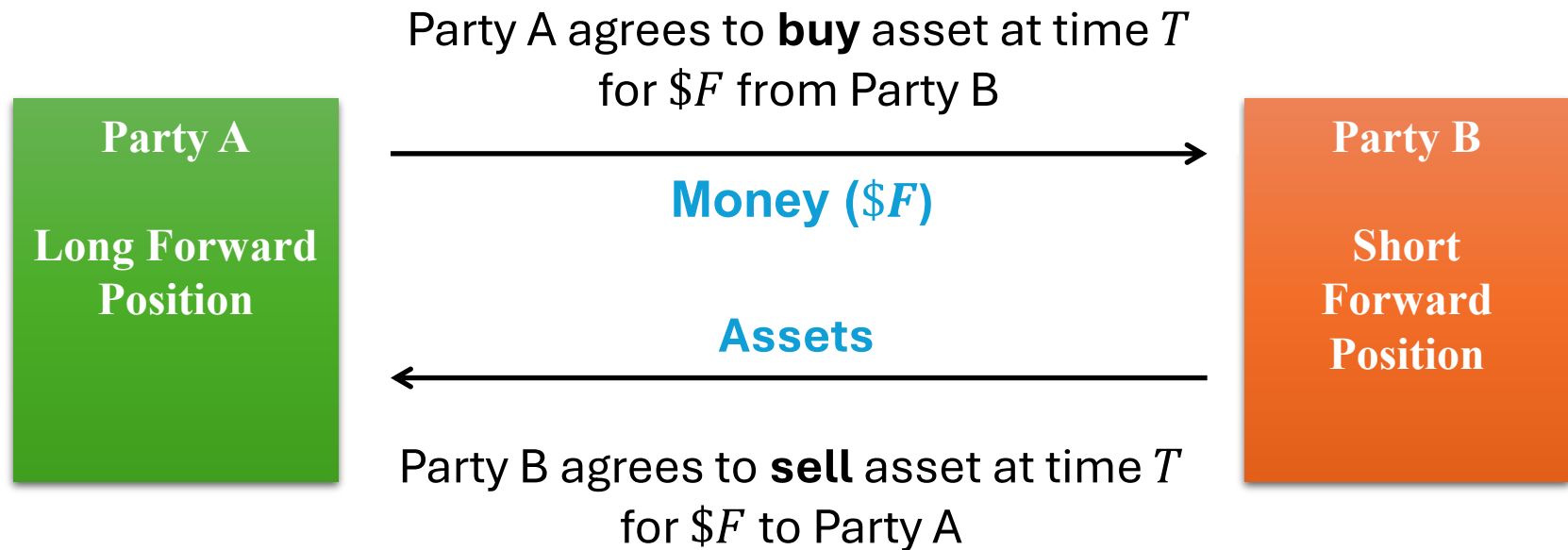
Forward Contracts

Spot and Forward

- Spot = today
- Forward = in the future
- A **spot contract** is an agreement to buy or sell an asset **today** at the **current** market (or “spot”) price S_0 .
 - These are quite common in our life; we do this every day.
- A **forward contract** is an agreement to buy or sell an asset on a **specified future** date at a **specified** price.
- Notation:
 - F – the delivery price of the forward contract
 - T – time remaining until delivery (time 0 is today).

Always two parties to a forward contract

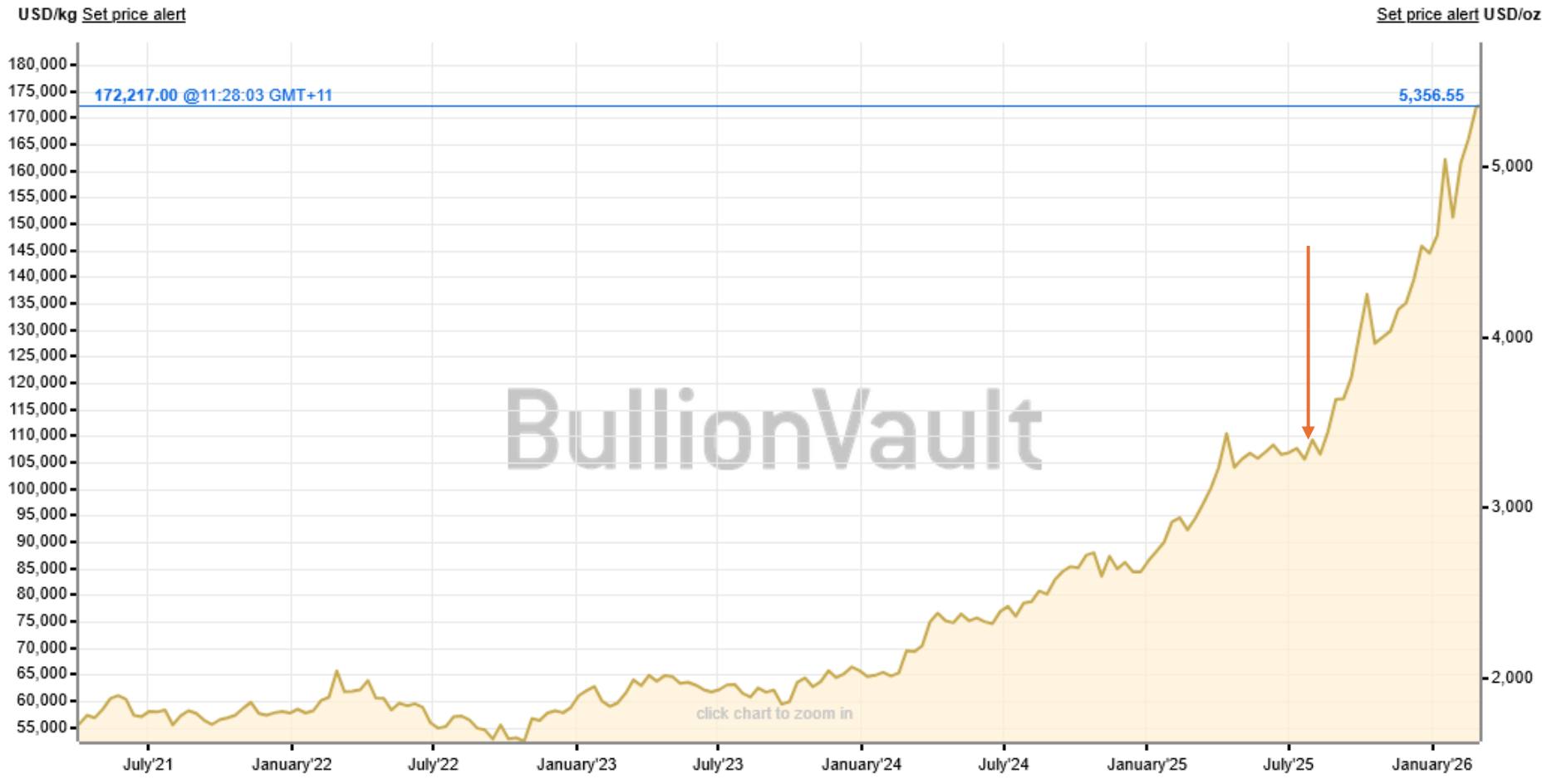
- **Long forward position**: the party who takes a **long** forward position is contracted to **buy** the underlying asset.
- **Short forward position**: the party who takes a **short** forward position is contracted to **sell** the underlying asset.



Speculating with forward contracts

- To illustrate how a forward contract works, consider an example where a forward is used for **speculation**:
 - Back in 2018, the spot price of gold was around USD 1,200 per ounce.
 - It rose sharply in early 2020 to over \$2,000 as covid-induced fear hit the financial markets. Investors often regard gold as a “safe haven” in troubled times.
 - It rose sharply again in Feb-2022 after Russia invaded Ukraine.
 - Gold rose ~75% since the beginning of 2024, and >60% since the beginning of 2025.
 - **As of 28/7/2025, gold price is USD 3,335 per ounce.**
 - (As of 2/3/2026b 11:28, gold price is USD 5,363 per ounce)

Spot price of gold (last 5 years)

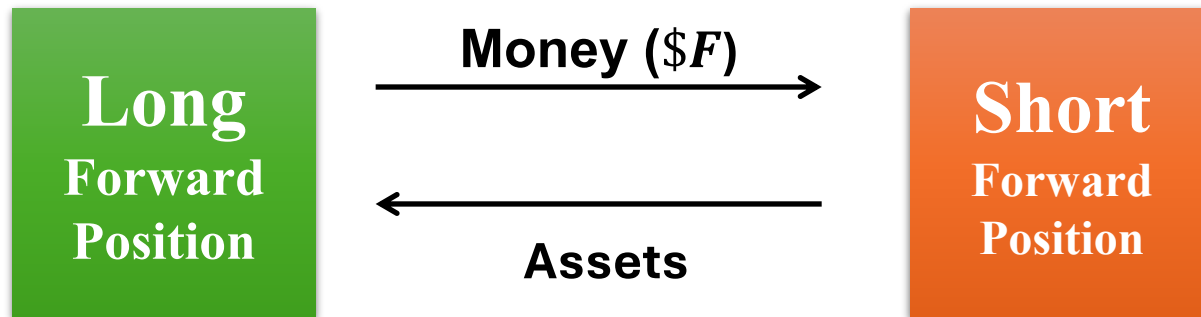


Source: <https://www.bullionvault.com/gold-price-chart.do>

Speculating with forward contracts

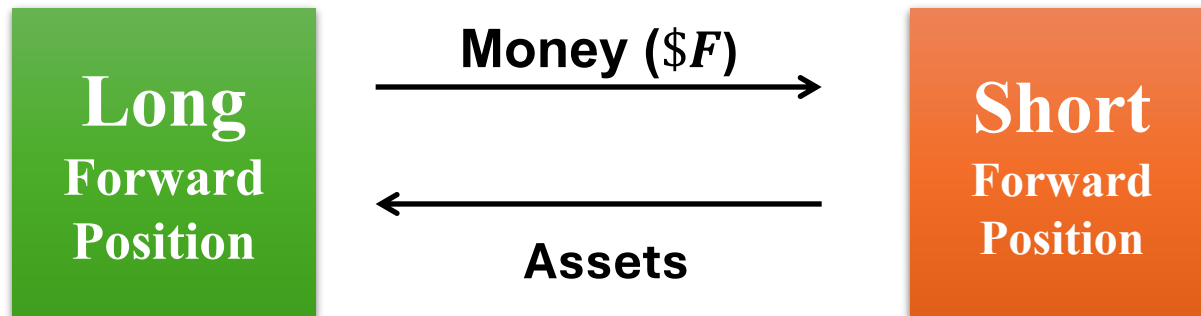
- Suppose that you are on **28/7/2025** (gold = **USD 3,335/oz**).
- You speculate that gold price will fall because inflation fears will calm down
 - You predict gold price will fall to **USD ~3,200/oz by the end of 2025**.
- You want to **speculate** on that (=make money from that prediction).
- You contact your broker (say, your bank) and learn that the **forward price (F) for delivery of gold in Dec-2025 is USD 3,390**.
 - We will learn how banks set the price (F) for a forward contract.
- **Q: Should you long or short forward contracts?**

Speculating with forward contracts



- The forward contract allows me to buy/sell gold on Dec-2025 at a price of $F=3,390$ USD/oz:
 - I am predicting that price will be $3,200 < F=3,390$.
- I would love to have a contract that allows me to sell gold for USD 3,390!
- Hence, I enter a **short** forward contract for Dec-2025 delivery of gold at $F = \text{USD } 3,390$.
 - Let's assume that I short a total size of **1,000 ounces of gold**,

Future Contract at the expiration date



- Short forward contract $\Rightarrow$ I am obliged to deliver 1,000 ounces of gold at the contract price of USD 3,390/oz on 29/Dec/2025.
- On 29/12/25 I will
 - Go to the spot market and buy 1,000 ounces at the **spot price**.
 - Deliver these 1,000 ounces to my counterparty for **USD 3,390/oz**.
- Both me and my counterparty are obliged to fulfill our part.

Expected profit - a good scenario

- Suppose that your prediction was pretty much correct and the **spot price of gold has fallen to USD 3,210** by December.

Action	USD
Buy 1,000 ounces of Gold at $S = 3,210$	-3,210,000
Sell (deliver) 1,000 ounces of gold at $F = 3,390$	+3,390,000
Total Profit	180,000

- When you short a forward, you **make profit** when the spot price at the delivery date is **lower than** the forward price.

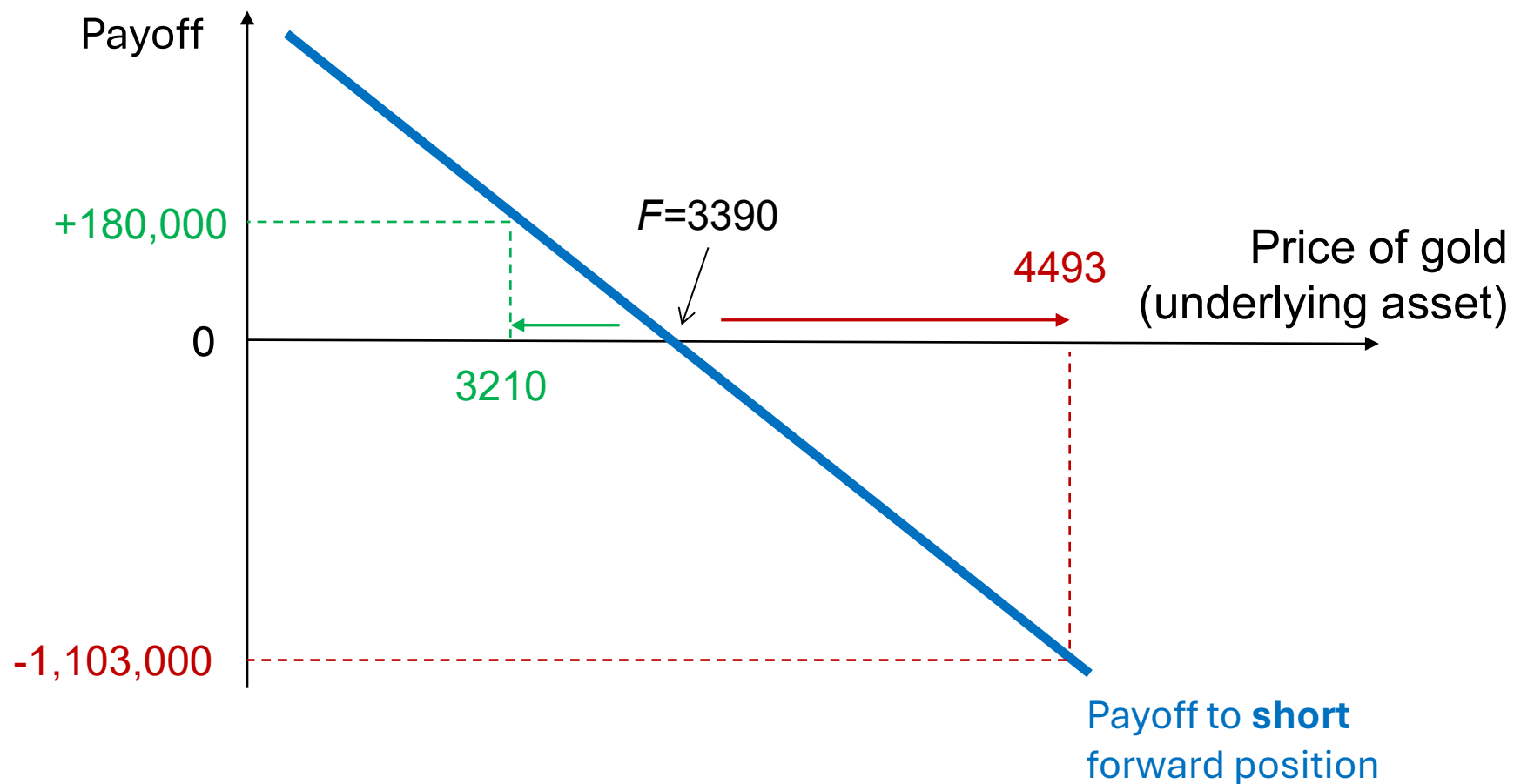
Expected profit - a bad scenario

- Suppose that the **spot price of gold has risen to USD 4,493** by December (**that's what actually happened**).

Action	USD
Buy 1,000 ounces of Gold at $S = 3,850$	-4,493,000
Sell (deliver) 1,000 ounces of gold at $F = 3,390$	+3,390,000
Total Profit (loss)	-1,103,000

- When you short a forward, you **lose money** when the spot price at the delivery date is **higher** than the forward price.

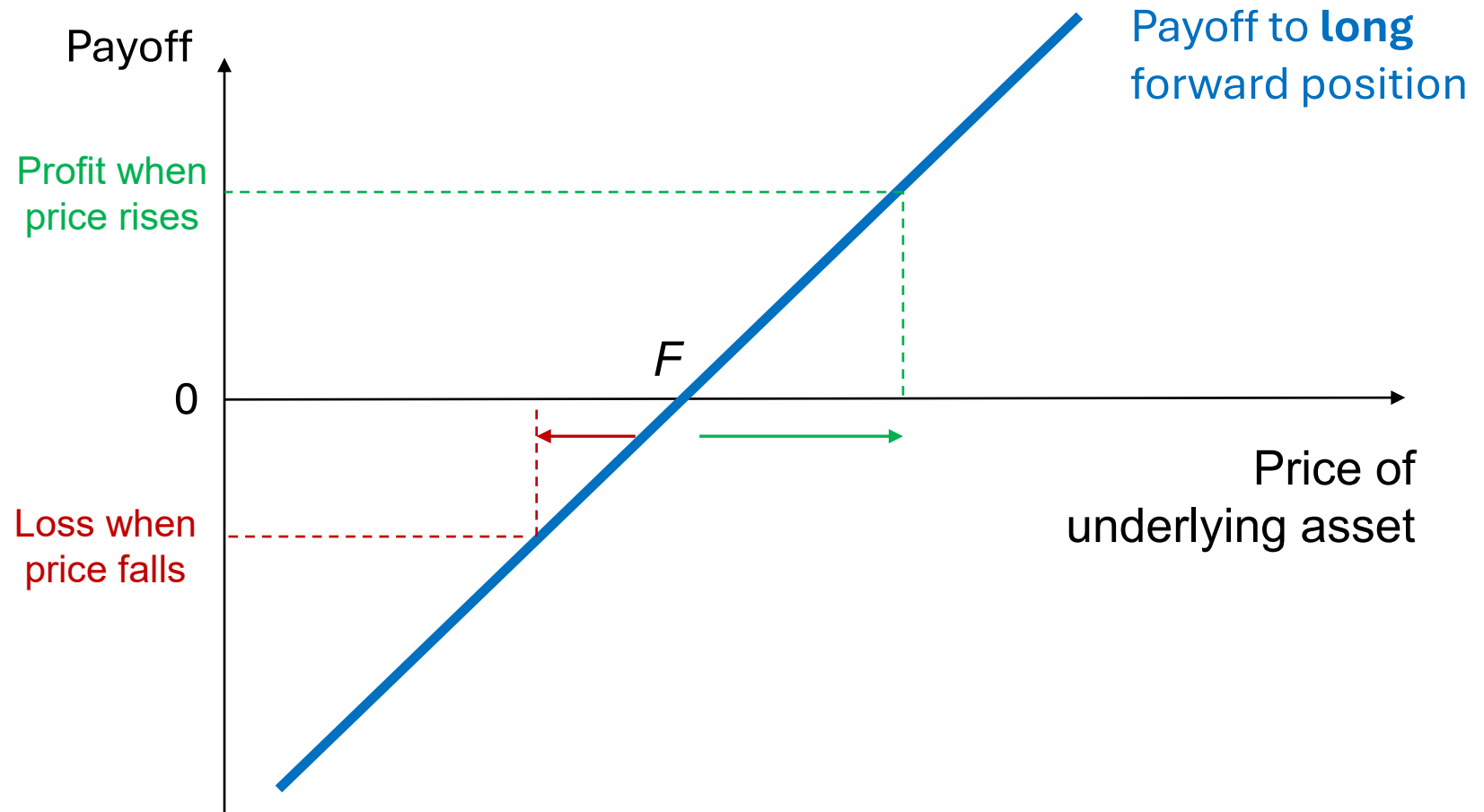
Payoff diagram: Short forward position



Payoff diagram: Long forward position

- You will soon see some examples involving long forward contracts.
- With a **long** forward position:
 - You **profit** on expiration when the price of the underlying asset $>$ the forward price, and
 - You **lose** on expiration when the price of the underlying asset $<$ the forward price.
- This can be depicted using the **payoff diagram** for a long forward position.

Payoff diagram: Long forward position

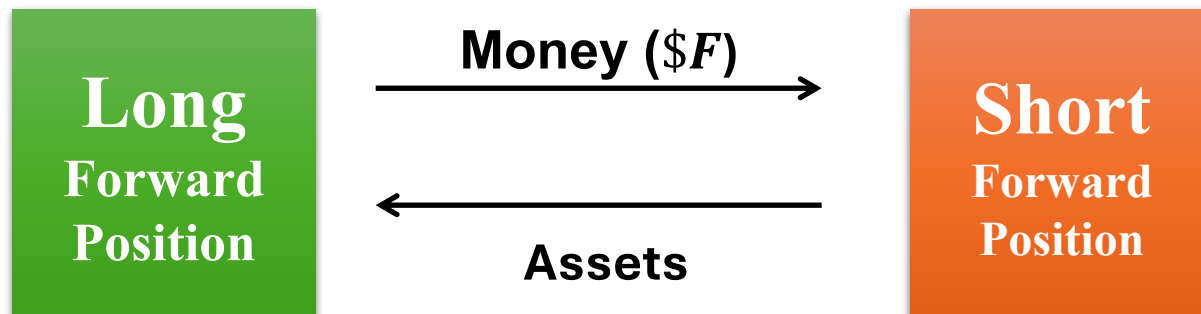


Where do forward contracts trade?

- Forward contracts are traded in an **over-the-counter (OTC)** market:
 - In OTC you trade directly with dealers, who communicate with each other by phone and/or computer.
 - It is a network of dealers working at financial institutions, corporations or fund managers.
- Financial institutions often act as “market makers”:
 - When a customer wants to enter a long forward contract, the institution is prepared to take the short side. And vice versa.
 - they quote **bid** prices (price at which they are prepared to go long).
 - and quote **offer** prices (price at which they are prepared to go short).

Defining characteristics of forward contracts

- Because they are arranged directly between two parties, forward contracts can be **highly customised** (wrt size, timing, delivery, underlying asset).
- Each party bears some **counterparty/credit risk**:
 - The risk that the counterparty defaults and does not fulfill its obligations.
 - Parties usually demand collateral from each other
 - Some OTC markets have adopted a CCP (Central Counterparty) to solve this.



Futures Contracts

Futures Contracts

- Just like a forward contract, a **futures contract** is also an agreement to buy or sell an asset at a **specified** price on a **specified future** date.
- Same notation:
 - F – the delivery price of the forward contract
 - T – time remaining until delivery (time 0 is today).
- In terms of their outcomes and the ways in which they are used, **there is little difference between forwards and futures**.
 - They both have long (short) parties who agree to buy (sell) the underlying asset at a fixed price at a given time.
 - They can both be used in similar ways to hedge risk or speculate.

Differences between Forwards and Futures

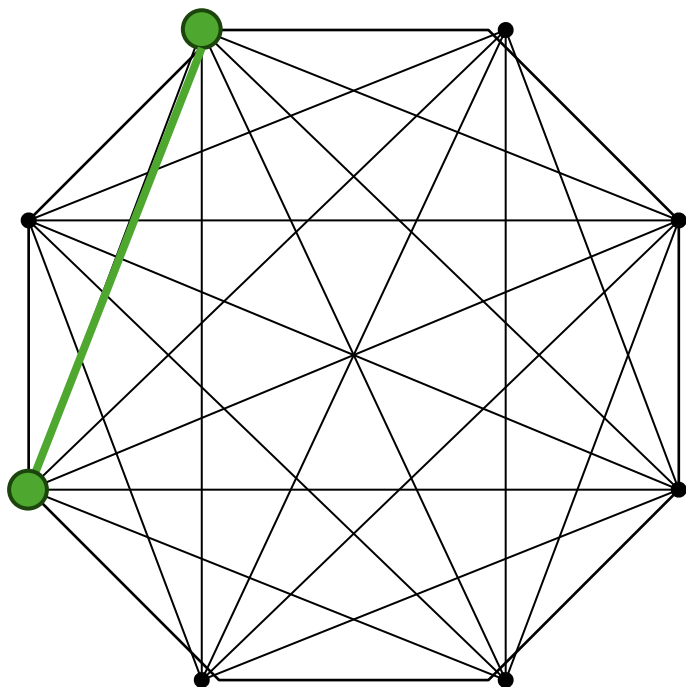
- Futures contracts trade on **organized exchanges**.
- The contracting parties do **not** face **credit risk**.
- Futures contracts are **highly standardised**.
- Futures contracts are “**marked to the market**” daily.
- Futures contracts are often **cash settled**, with no physical delivery of the underlying asset.

Examples of large futures exchanges

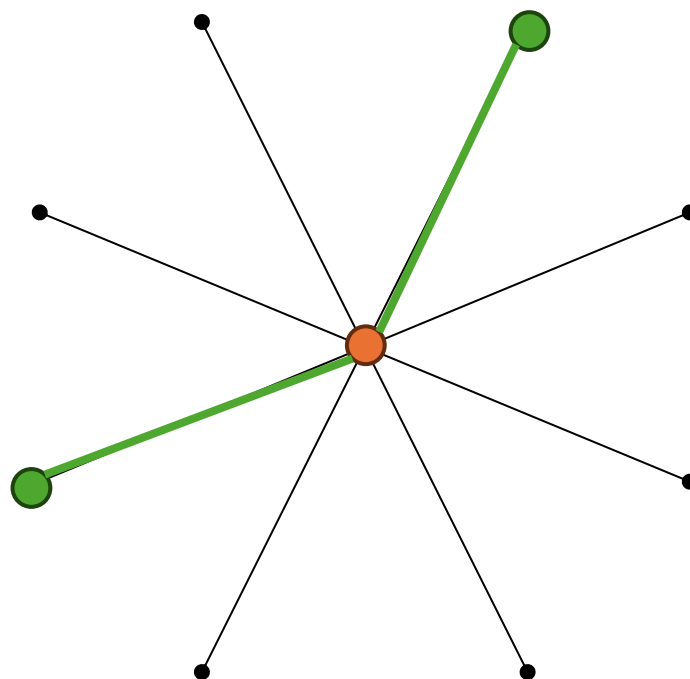
- CME Group (www.cmegroup.com)
 - New York Mercantile Exchange (NYME)
 - Chicago Board of Exchange (CBOE)
- The Intercontinental Exchange (ICE) (www.theice.com)
- Eurex (www.eurexexchange.com),
- B3, Brazil (www.b3.com.br)
- National Stock Exchange of India (www.nse-india.com)
- China Financial Futures Exchange (www.cffex.com.cn)
- Tokyo Financial Exchange (www.tfx.co.jp). (See the table at the end of this book for a more
- Australian Securities Exchange (ASX) – ASX Trade24

Organized exchange trading

OTC market with bilateral agreements between two parties



An organized exchanges (or an OTC market with CCP), where all agreements go through central clearing house



Source: Hull (2018, Figure 2.2)

Organised exchange trading

- Some big advantages of having the contracts trade on an exchange:
 - No effort required to find a counterparty.
 - No credit/default risk.
 - The exchange's clearing house stands between the parties and deals with credit risk by requiring margin.
 - Margin is calculated daily.

Standardised contract specifications

- Each exchange has a closed list of standardized products that are traded.
- The specific terms of a futures contracts are highly standardized with respect to:
 - **Underlying asset:** precisely what asset must be delivered and what its quality must be.
 - **Contract size:** the amount of the underlying asset that has to be delivered under one contract.
 - **Delivery date:** which dates the contract expires in.
 - **Delivery terms:** if physical delivery is possible, where, when and how.

Example - Gold Futures

- <https://au.investing.com/commodities/gold?cid=1213656>
- <https://www.cmegroup.com/markets/metals/precious/gold/contractSpecs.html>

Prev. Close	3,392.5
Open	3,390.8
Day's Range	3,384.6 - 3,396.1
52 wk Range	2,442.3 - 3,544.2
Volume	6,511
1-Year Change	37.99%
Month	Dec 25
Contract Size	100 Troy Ounces

Settlement Day	29/12/2025
Settlement Type	Physical
Last Rollover Day	25/11/2025
Tick Size	0.1
Tick Value	10
Base Symbol	ZG
Point Value	1 = \$100
Months	GJMQVZ

Example of Standardisation in commodities

(These are just examples. You do not need to remember the details)

- **NYSE Orange Juice Futures** - US Grade A, with Brix value of not less than 57 degrees, having a Brix value to acid ratio of not less than 13 to 1 nor more than 19 to 1, with factors of colour and flavour each scoring 37 points or higher and 19 for defects, with a minimum score 94.
- **CME Lumber Futures** - Delivery on track and shall either be unitized in double-doored boxcars or, at no additional cost to the buyer, each unit shall be individually paper-wrapped and loaded on flatcars. Par delivery of hem-fir in California, Idaho, Montana, Nevada, Oregon, and Washington, and in the province of British Columbia.

CBOT corn futures contract specs

Contract Size	5,000 bushels (~ 127 Metric Tons)
Deliverable Grade	#2 Yellow at contract Price, #1 Yellow at a 1.5 cent/bushel premium #3 Yellow at a 1.5 cent/bushel discount
Pricing Unit	Cents per bushel
Tick Size (minimum fluctuation)	1/4 of one cent per bushel (\$12.50 per contract)
Contract Months/Symbols	March (H), May (K), July (N), September (U) & December (Z)
Trading Hours	CME Globex (Electronic Platform) 6:00 pm - 7:15 am and 9:30 am - 1:15 pm central time, Sunday - Friday Central Time Open Outcry (Trading Floor) 9:30 am - 1:15 pm Monday - Friday Central Time
Daily Price Limit	\$0.30 per bushel expandable to \$0.45 and then to \$0.70 when the market closes at limit bid or limit offer. There shall be no price limits on the current month contract on or after the second business day preceding the first day of the delivery month.
Settlement Procedure	Physical Delivery
Last Trade Date	The business day prior to the 15th calendar day of the contract month.
Last Delivery Date	Second business day following the last trading day of the delivery month.
Product Ticker Symbols	CME Globex (Electronic Platform) ZC C=Clearing Open Outcry (Trading Floor) C
Exchange Rule	These contracts are listed with, and subject to, the rules and regulations of CBOT.

Margin Requirements and Mark to market

- Futures contracts are **marked to the market** daily.
- When you open a futures position (long or short), you must lodge an **initial margin** (i.e., a deposit) with the exchange.
 - At the end of each day, the exchange calculates your gains/losses given a daily closing price. These are added to (subtracted from) your margin account.
 - If your margin balance gets too low, the exchange makes a **margin call** (i.e., requires you to deposit more funds).
- **Margins are incredibly important in futures trading, but we do not discuss them in detail in this course.**

Settlement procedures for futures

- Once you have entered a futures position (either long or short), you may
 1. hold that position until the expiry/delivery date, or
 2. “close-out” your position prior to expiry.
- Let’s consider how each of these plays out.

Holding until expiry – physical delivery

- Some futures contracts allow physical delivery of the underlying asset on the expiry date.
 - The contract specifications clearly articulate where, when and how the delivery must take place.
 - For example, in the case of the ICE frozen concentrate orange juice contract, delivery is to exchange-licensed warehouses in Florida, New Jersey, or Delaware.
- **In such cases, the futures contract is very similar to a forward contract.**
- In practice, it is **rare** that futures contracts result in physical delivery:
 - Most futures are closed out prior to expiry [see below].
 - They have served their purpose (hedging or speculation).
 - Physical delivery is more trouble than it is worth.

Holding until expiry – Cash settlement

- Some futures contracts cannot be physically delivered.
 - Stock index futures are on stock portfolios that are difficult to build.
 - Weather futures, traded on the CME, are based on indices that measure temp. deviations (Heating/Cooling Degree Days). No physical asset.
- Instead, these futures are **cash settled**.
- The profit/loss of the trader's position is calculated using the settlement price at the date of expiry
 - Which is based on the spot price of the underlying asset.
- The exchange transfers money to/from the trader's account according to her profit/loss.
 - Note the exchange does the opposite transfer to the other side of the contract, and thus always remains balanced.

Cash settlement at expiry – Example

- On January 22nd, 2024, we enter 2 **long** Sep-24 SPI200 futures contracts at 6677.
- SPI200 futures have a standard contract size of \$25 times the index.

22 January 2024

EQUITY FUTURES & OPTIONS

		Prev price	Opening Trade	High	Low	Settlement Price	Change	Volume	Prev O/P
ASX SPI 200 Index Future (AP)									
Jul	24	6771.0	6750.0	6772.0	6750	6771	1	94	2589
Aug	24	6756.0	6757.0	6763.0	6757	6756	7	1	11
Sep	24	6677.0	6678.0	6678.0	6660	6677	-15	35769	248692
Dec	24	6684.0	-	-	-	6692	8	1	593
Jun	25	6617.0	-	-	-	6625	8	1	2
Total									
Volume	35866								Prev O/P: 251887

Cash settlement at expiry – Example

- On January 22nd, 2024, we enter 2 **long** Sep-24 SPI200 futures contracts at 6677.
 - SPI200 futures have a standard contract size of **\$25 times the index**.
- Suppose we hold the contract until expiry on Sep 30, 2024.
- Our long position implies we **gain** if the index has increased and **lose** if it decreased.
 - This gain/loss is Australia **\$25 for each index point**.
- Intuition (may help some and confuse others):
 - Our payoff from each contract is **as-if** we
 - pay \$25 times $F = 6,677$
 - “Obtain” the SPI200 and sell it for \$25 times the index at the time of expiry.

Cash settlement at expiry – Example

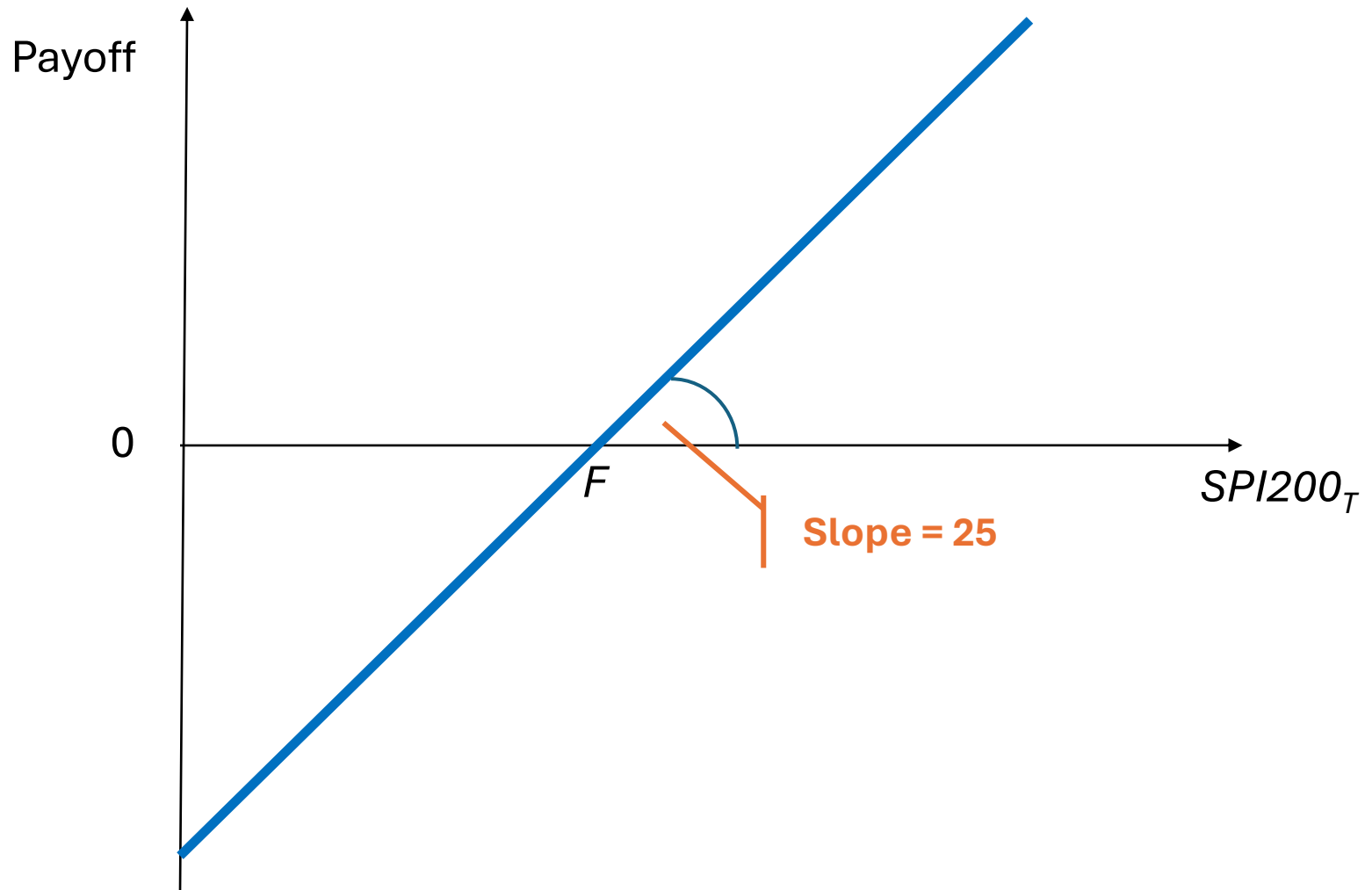
- On January 22nd, 2024, we enter 2 **long** Sep-24 SPI200 futures contracts at 6677.
 - SPI200 futures have a standard contract size of \$25 times the index.
- Suppose we hold the contract until expiry on Sep 30, 2024.
- If the SPI200 index is 7,000 at expiry (30/9/24), then we make a **positive** profit of

$$\$16,150 = 2 \text{ contracts} \times (7,000 - 6,677) \times \$25$$

- If the SPI200 index is 6,100 at expiry (30/9/24), then we make a **negative** profit (loss) of

$$-\$28,850 = 2 \text{ contracts} \times (6,100 - 6,677) \times \$25$$

Payoff diagram: long futures position (at expiry)



Closing-out prior to expiry

- All futures contracts can be “closed-out” prior to expiry.
- This is done by entering a new futures position **equal in magnitude but opposite in direction** to the original futures position:
 - If the opening futures positions is **long**, it is closed out by entering a **short** position in the same futures contract.
 - If the opening futures positions is **short**, it is closed out by entering a **long** position in the same futures contract.
- The **profit** from closing out a position early depends on the **difference** between the future prices when the position is **opened** and when it is **closed**.

Closing-out prior to expiry – Example

- On **January 22nd, 2024**, we enter a **long position** on (buy) a single futures contract of Eastern Australian Wheat **Jan '25** at $F = \$422/\text{tonne}$.
- Wheat futures have a standard contract size of 20 metric tonnes.

22 January 2024

AGRICULTURAL FUTURES									
		Prev price	Opening Trade	High	Low	Settlement Price	Change	Volume	Prev O/P
Eastern Australia Wheat (WM)									
Jul	24	414.3	-	-	-	408.5	-5.8	62	577
Sep	24	415	-	-	-	415	0	62	1309
Jan	25	422	-	-	-	422	0	150	10875
Jan	26	432.5	-	-	-	432.5	0	5	180
Total									
Volume:	279	Prev O/P: 14250							

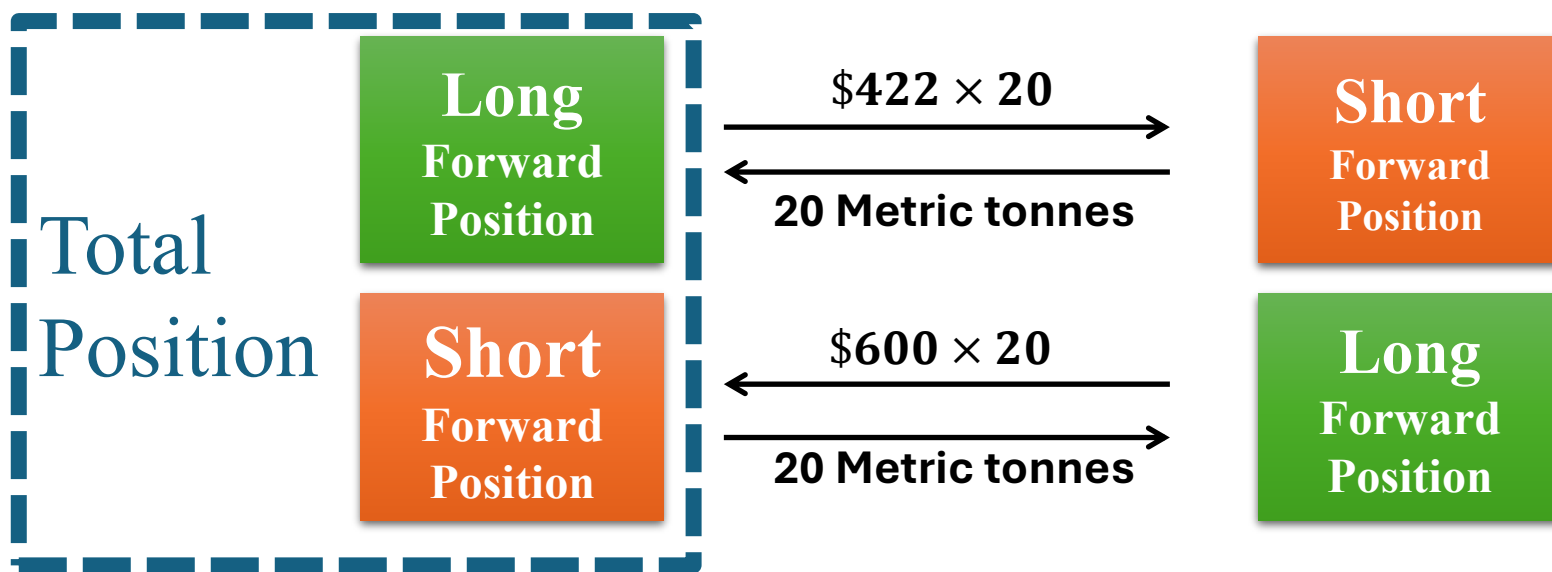
Closing-out prior to expiry – Example

- On **January 22nd, 2024**, we enter a **long position** on (buy) a single futures contract of Eastern Australian Wheat **Jan '25** at $F = \$422/\text{tonne}$.
 - Wheat futures have a standard contract size of 20 metric tonnes.
- On **March 1st, 2024**, we close-out our position by **selling** a single futures contract of Eastern Australian Wheat Jan '25.
- If the wheat futures price on 1/3 is $F = 600$ when we close:
 - We make a profit of $\$3,560 = (600 - 422) \times 20$ tonnes
- If the wheat futures price on 1/3 is $F = 250$ when we close:
 - We make a loss: Profit = **$-\$3,440$** = $(250 - 422) \times 20$ tonnes

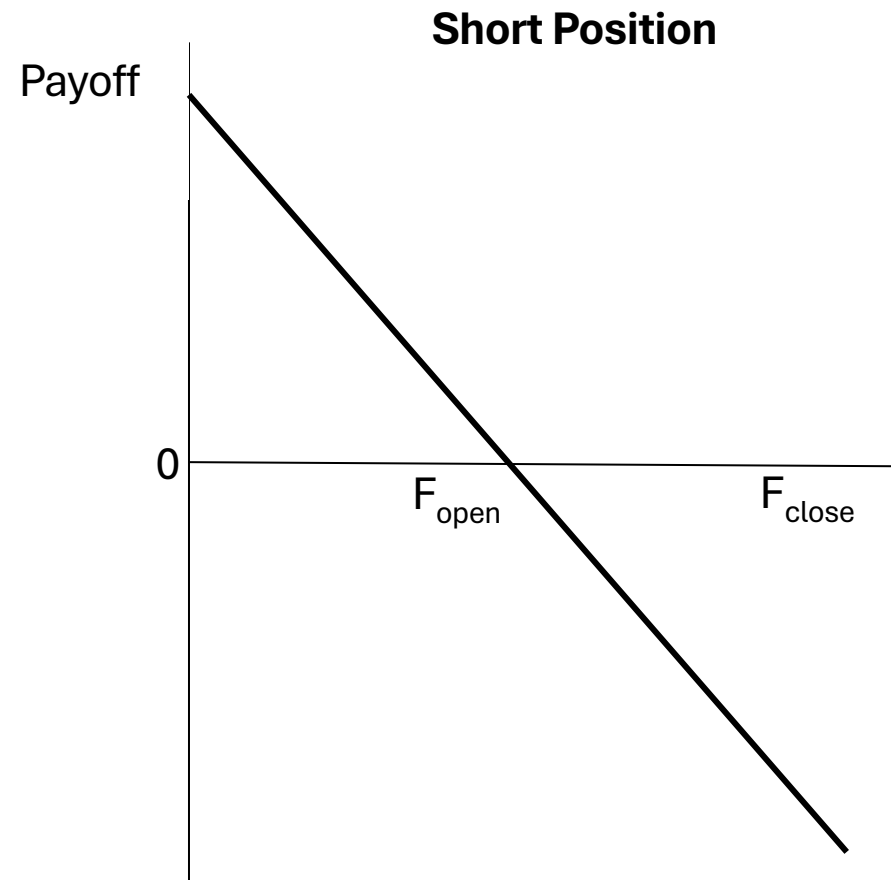
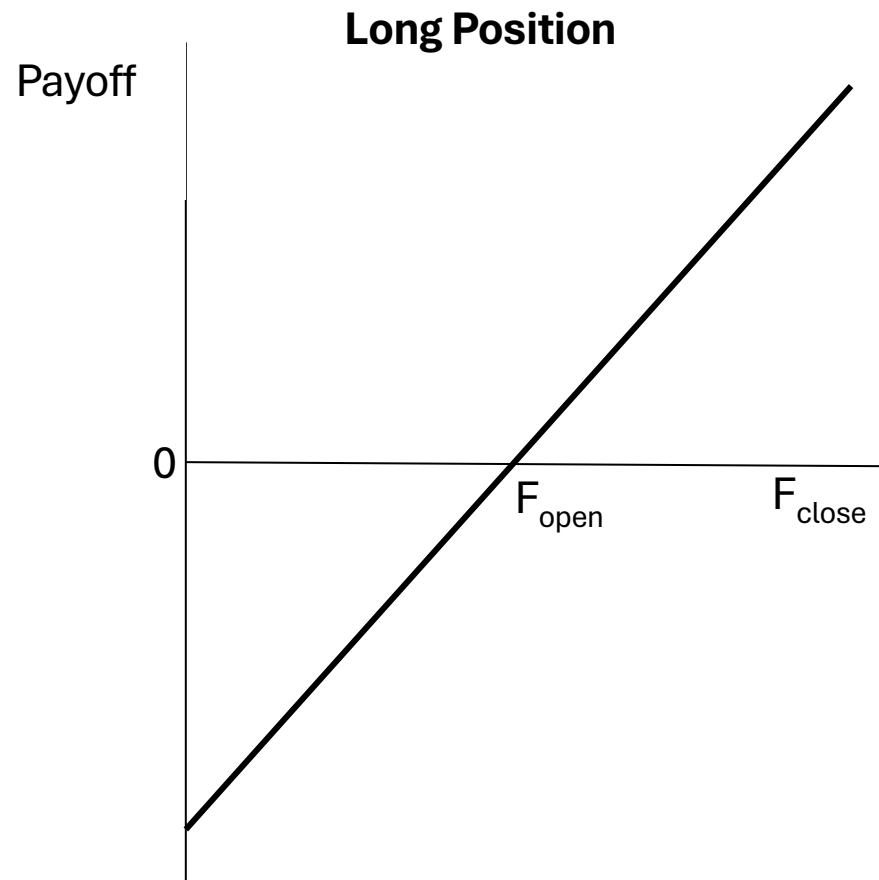
Closing-out prior to expiry – Example

A more detailed explanation of the payoff from closing-out when the Price is \$600.

- **Long** 1 futures contract of Eastern Australian Wheat Jan '25 at $F = \$422/\text{tonne}$.
- **Short** 1 futures contract of Eastern Australian Wheat Jan '25 at $F = \$600/\text{tonne}$.
- Wheat futures have a standard contract size of 20 metric tonnes.



Payoff diagrams for futures contracts



Futures contracts in Australia (ASX)

- Equities
 - share price index (SPI200) futures
- Interest rates
 - 30-day interbank cash rate
 - 90-day bank accepted bill (BAB) futures
 - 3-year, 10-year and 20-year bond futures
- Agricultural:
 - Eastern Australia wheat futures, WA wheat futures
 - Eastern Australia feed barley futures
- Energy futures: some base load electricity contracts

Futures trading on CME

- US exchanges offer trading on a much wider variety of underlying assets:
 - **Equity Indexes** – S&P 500, Nasdaq-100, Dow Jones, Russell (standard, E-mini, Micro)
 - **Interest Rates** – U.S. Treasuries and benchmark rate futures
 - **Foreign Exchange** – Major, minor, and emerging-market currency futures
 - **Energy** – Crude oil, natural gas, refined products
 - **Metals** – Precious (gold, silver) and industrial (copper)
 - **Agriculture** – Grains, oilseeds, livestock, dairy, and softs
 - **Cryptocurrency** – Bitcoin and Ether futures
- See <http://www.cmegroup.com/trading/products/>

Speculating with futures → sport bets?

- Future contracts are not limited to asset prices.
 - Weather futures, traded on the CME, are based on indices that measure temperature deviations (Heating/Cooling Degree Days).
 - These contracts are primarily used for hedging purposes in agriculture and energy sectors, though they can also be used for speculation.
- In theory, future contracts could be created around the score difference of a sports match.
- The distinction between speculation and betting is often unclear and depends on regulatory frameworks.
 - Kalshi, a prediction market, operates as a “market for trading event contracts” and is regulated by the CFTC in the US.
 - This regulatory status enables it to avoid state gambling laws.

Comparison of forwards and futures

	Forwards	Futures
Traded where?	Over the counter	On exchanges
Terms of contract are	Tailored to your needs	Highly standardised
Delivery date is	Tailored to your needs	A range of standard delivery months
Settlement occurs	At expiry	Marked to market daily
What happens to the underlying asset?	Often physically delivered	Contract is usually closed out prior to expiry
Any credit risk?	Yes, some	None

Key takeaways from this lecture

- Forward contracts:
 - What are they? How do they work? Defining characteristics.
 - Difference between long and short forward positions.
 - What are your obligations under long and short forward contracts?
 - Being able to calculate the profit/loss on forward positions.
- Futures contracts:
 - What are they? How do they work? Defining characteristics.
 - Difference between long and short futures positions.
 - What are your obligations under long and short futures contracts?
 - Being able to calculate the profit/loss on futures positions.
 - Closing-out vs. holding till expiry. Physical delivery vs. cash settlement.